

Esercizio derivazioni

21 January 2023 15:40

1. Considerate la grammatica $G = (\{S, A, B\}, \{a, b\}, \mathcal{P}, S)$ dove $\mathcal{P}$ contiene le produzioni $S \rightarrow aSa, S \rightarrow bSb, S \rightarrow aAb, A \rightarrow aAb, A \rightarrow aBb, A \rightarrow BA, A \rightarrow AB, A \rightarrow b, B \rightarrow bAa, B \rightarrow bAb, B \rightarrow b$ e $B \rightarrow a$.

- Scrivere una derivazione per la parola $abaaabbbba$,
- Scrivere una derivazione per la parola $abbaababb$,
- Scrivere una derivazione per la parola $abababbbba$,
- Scrivere una derivazione per la parola $ababbbabb$,
- Scrivere una derivazione per la parola $abbbabbab$,
- Scrivere una derivazione per la parola $bbaaabbbb$,
- Scrivere una derivazione per la parola $baaaabbbab$,
- Scrivere una derivazione per la parola $aaaaabbaa$,
- Scrivere una derivazione per la parola $aabbbabb$,
- Scrivere una derivazione per la parola $abbbbbb$.

(d) ababbbabb

S
 $\downarrow S \rightarrow aAb$
 aAb
 $\downarrow A \rightarrow BA$
 $aBAb$
 $\downarrow B \rightarrow b, A \rightarrow aBb$
 $ababbb$
 $\downarrow B \rightarrow bAa$
 $ababbaa$
 $\downarrow A \rightarrow AB$
 $ababbaab$
 $\downarrow A \rightarrow b, B \rightarrow b$
 $ababbbabb$

$\left\{ \begin{array}{l} S \rightarrow aSa \mid bSb \mid aAb \\ A \rightarrow aAb \mid aBb \mid BA \mid AB \mid b \\ B \rightarrow bAa \mid bAb \mid b \mid a \end{array} \right.$

$aAb - BA - b - aBb - bAa - AB - b - b$

(e) abbbabbab

S
 $\downarrow S \rightarrow aAb$
 aAb
 $\downarrow A \rightarrow AB$
 $aABb$
 $\downarrow A \rightarrow b, B \rightarrow bAa$
 $abbbAa$
 $\downarrow A \rightarrow BA$
 $abbbBAa$
 $\downarrow B \rightarrow bAb, A \rightarrow b$
 $abbbAbab$
 $\downarrow A \rightarrow BA$

$\left\{ \begin{array}{l} S \rightarrow aSa \mid bSb \mid aAb \\ A \rightarrow aAb \mid aBb \mid BA \mid AB \mid b \\ B \rightarrow bAa \mid bAb \mid b \mid a \end{array} \right.$

$a b b b A b a b$
 $\downarrow A \rightarrow BA$
 $a b b b B A b a b$
 $\downarrow B \rightarrow a, A \rightarrow b$
 $a b b b a b b a b$

(f)

$b b a a a b b b b$

S
 $\downarrow S \rightarrow b S b$
 $b S b$
 $\downarrow S \rightarrow b S b$
 $b b S b b$
 $\downarrow S \rightarrow a A b$
 $b b a A b b b$
 $\downarrow A \rightarrow a B b$
 $b b a a B b b b b$
 $\downarrow B \rightarrow a$
 $b b a a a b b b b$

$\left\{ \begin{array}{l} S \rightarrow a S a \mid b S b \mid a A b \\ A \rightarrow a A b \mid a B b \mid B A \mid A B \mid b \\ B \rightarrow b A a \mid b A b \mid b \mid a \end{array} \right.$

(g)

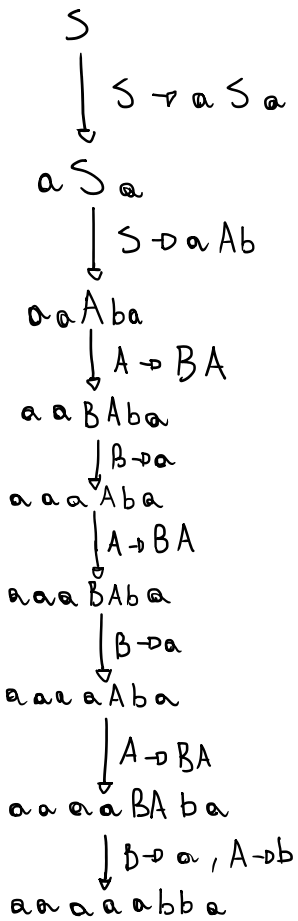
$b a a a a b b a b$

S
 $\downarrow S \rightarrow b S b$
 $b S b$
 $\downarrow S \rightarrow a S a$
 $b a S a b$
 $\downarrow S \rightarrow a A b$
 $b a a A b a b$
 $\downarrow A \rightarrow a A b$
 $b a a a B b b a b$
 $\downarrow B \rightarrow a$
 $b a a a a b b a b$

$\left\{ \begin{array}{l} S \rightarrow a S a \mid b S b \mid a A b \\ A \rightarrow a A b \mid a B b \mid B A \mid A B \mid b \\ B \rightarrow b A a \mid b A b \mid b \mid a \end{array} \right.$

h

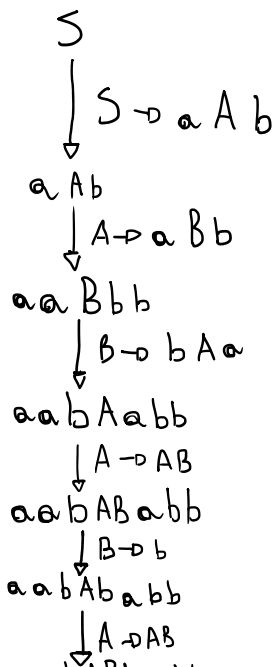
aaaaabba



$$\begin{cases}
 S \rightarrow aSa \mid bSb \mid aAb \\
 A \rightarrow aAb \mid aBb \mid BA \mid AB \mid b \\
 B \rightarrow bAa \mid bAb \mid b \mid a
 \end{cases}$$

i

aabbbabb



$$\begin{cases}
 S \rightarrow aSa \mid bSb \mid aAb \\
 A \rightarrow aAb \mid aBb \mid BA \mid AB \mid b \\
 B \rightarrow bAa \mid bAb \mid b \mid a
 \end{cases}$$

$a a b A b b a b b$
 $\downarrow A \rightarrow b, B \rightarrow b$
 $a a b b b b a b b$

(5)

$a b b b b b b b$

S
 $\downarrow A \rightarrow b$
 $a A b$
 $\downarrow A \rightarrow AB$
 $a A B b$
 $\downarrow B \rightarrow b A b$
 $a A b A b b$
 $\downarrow A_1 \rightarrow b, A_2 \rightarrow AB$
 $a b b A B b b$
 $\downarrow B \rightarrow b A b$
 $a b b A b A b b b$
 $\downarrow A_{1,2} \rightarrow b$
 $a b b b b b b b$

5. Considerate la grammatica $G = (\{S, A\}, \{a, b\}, \mathcal{P}, S)$ dove $\mathcal{P}$ contiene le produzioni $S \rightarrow aSb$, $S \rightarrow aAb$, $A \rightarrow aaA$ e $A \rightarrow aa$. Dite se il linguaggio generato dalle produzioni è quello delle parole in cui il numero di a è uguale al numero di b più un numero pari qualsiasi maggiore di 0. La risposta non può essere un semplice *si* o *no* ma deve essere argomentata.

$S \xrightarrow{S \rightarrow aAb} a A b \xrightarrow{A \rightarrow aa} a a a b$

L'enunciato è vero, in quanto le regole di produzione permettono solo di aggiungere a e b insieme contemporaneamente sfruttando il simbolo iniziale S , mentre per raggiungere lo stato finale della parola si deve obbligatoriamente aggiungere un numero pari di aa .

6. Considerate la grammatica $G = (\{S, A, B\}, \{a, b\}, \mathcal{P}, S)$ dove $\mathcal{P}$ contiene le produzioni $S \rightarrow aAbBa$, $S \rightarrow aAb$, $A \rightarrow ab$, $B \rightarrow bBa$ e $B \rightarrow b$. Dite se il linguaggio generato dalle produzioni è quello delle parole in cui il numero di a è uguale al numero di b più uno e se una b è preceduta da una a allora prima della a ci sono solo a , e se una b è seguita da una a allora dopo la a ci sono solo a . La risposta non può essere un semplice *si* o *no* ma deve essere argomentata.

$S \xrightarrow{1} a A b B a \xrightarrow{2} a A b$
 $A \rightarrow ab$
 $B \rightarrow b B a \mid b$

$S \xrightarrow{S \rightarrow aAb} a A b \xrightarrow{A \rightarrow ab} a a b b$
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7. Considerate il seguente λ -termine $(\lambda x. \lambda y. xy)(\lambda z. z)(\lambda y. yy)$. Identificate i redexes del termine ed eseguite un passo di β -riduzione.

$$1 \quad (\lambda x. \lambda y. x y) (\lambda z. z) (\lambda y. y y) \quad 2$$

$$1 \quad (\lambda y. (\lambda z. z) y) (\lambda y. y y) \rightarrow (\lambda z. z) (\lambda y. y y) \rightarrow (\lambda y. y y)$$

$$2 \quad (\lambda x. \lambda y. x y) (\lambda y. y y)$$

8. Considerate il seguente λ -termine $(\lambda x. \lambda y. x x y)(\lambda z. z)(\lambda y. y)$. Identificate i redexes del termine ed eseguite un passo di β -riduzione.

$$1 \quad (\lambda x. \lambda y. x x y) (\lambda z. z) (\lambda y. y) \quad 2$$

$$1 \quad (\lambda y. (\lambda z. z) (\lambda z. z) y) (\lambda y. y)$$

$$2 \quad (\lambda x. \lambda y. x x y) (\lambda y. y)$$

9. Considerate il seguente λ -termine $(\lambda x. \lambda y. y x y)(\lambda z. z)(\lambda y. y)$. Identificate i redexes del termine ed eseguite un passo di β -riduzione.

$$1 \quad (\lambda x. \lambda y. y x y) (\lambda z. z) (\lambda y. y) \quad 2$$

$$1 \quad (\lambda y. y (\lambda z. z) y) (\lambda y. y)$$

$$2 \quad (\lambda x. \lambda y. y x y) (\lambda y. y)$$

10. Considerate il seguente λ -termine $(\lambda x. \lambda y. x y y)(\lambda z. z)(\lambda y. y)$. Identificate i redexes del termine ed eseguite un passo di β -riduzione.

$$1 \quad (\lambda x. \lambda y. x y y) (\lambda z. z) (\lambda y. y) \quad 2$$

$$1 \quad (\lambda y. (\lambda z. z) y y) (\lambda y. y)$$

$$2 \quad (\lambda x. \lambda y. x y y) (\lambda y. y)$$

11. Considerate il seguente λ -termine $(\lambda x. \lambda y. x v y)(\lambda z. w z)(\lambda y. y v)$. Quali sono le variabili libere del termine? Eseguite, se possibile, un passo di β -riduzione.

$$1 \quad (\lambda x. \lambda y. x v y) (\lambda z. w z) (\lambda y. y v) \quad 2 \quad \text{LIBERE} \rightarrow w, v$$

$$1 \quad (\lambda y. (\lambda z. w z) v y) (\lambda y. y v)$$

$$2 \quad (\lambda x. \lambda y. x v y) (w (\lambda y. y v))$$

12. Considerate il seguente λ -termine $(\lambda x. xvy)(\lambda z. wz)(\lambda y. yv)$. Quali sono le variabili libere del termine?
Eseguite, se possibile, un passo di β -riduzione.

$$1 \quad \underline{(\lambda x. xvy)(\lambda z. wz)(\lambda y. yv)} \quad 2 \quad \text{LIBERE} \rightarrow v, y, w$$

$$1 \quad ((\lambda z. wz)vy)(\lambda y. yv)$$

$$2 \quad (\lambda x. xvy) (w(\lambda y. yv))$$

13. Considerate il seguente λ -termine $(\lambda x. xy)(\lambda z. \lambda w. wz)(\lambda y. yv)$. Quali sono le variabili libere del termine?
Eseguite, se possibile, un passo di β -riduzione.

$$1 \quad \underline{(\lambda x. xy)(\lambda z. \lambda w. wz)(\lambda y. yv)} \quad 2 \quad \text{LIBERE} \rightarrow y, v$$

$$((\lambda z. \lambda w. wz)y)(\lambda y. yv)$$

$$(\lambda x. xy)(\lambda w. w(\lambda y. yv))$$

(a) $\{([], \perp, [\text{sum}(\text{apply}(\text{fun}(g, y, \text{sum}(y, 1)), 4), \text{apply}(\text{fun}(f, x, x), 2))])\}$

$[(\llbracket [], \perp, [\text{sum}(\text{apply}(\text{fun}(g, y, \text{sum}(y, 1)), 4), \text{apply}(\text{fun}(f, x, x), 2))\rrbracket)]$
 $[(\llbracket [], \perp, [\text{apply}(\text{fun}(g, y, \text{sum}(y, 1)), 4), \text{apply}(\text{fun}(f, x, x), 2), \text{print}(\text{sum})]\rrbracket)]$
 $[(\llbracket [], \perp, [\text{fun}(g, y, \text{sum}(y, 1)); 4; \text{app}; \text{apply}(\text{fun}(f, x, x), 2); \text{print}(\text{sum})]\rrbracket)]$
 $[(\llbracket 4, \text{fun}(g, y, \text{sum}(y, 1))\rrbracket, \perp, \text{app}; \text{apply}(\text{fun}(f, x, x), 2); \text{print}(\text{sum}))\rrbracket]$
 $[(\llbracket [], \perp, \text{fun}(g, y, \text{sum}(y, 1))\rrbracket; \text{y3}, [\text{sum}(y, 1)]; (\llbracket [], \perp, [\text{apply}(\text{fun}(f, x, x), 2); \text{print}(\text{sum})]\rrbracket)]$
 $[(\llbracket [], \perp, \text{fun}(g, y, \text{sum}(y, 1))\rrbracket; \text{y3}, [\text{y1}, \text{print}(\text{sum})]); (\llbracket [], \perp, [\text{apply}(\text{fun}(f, x, x), 2); \text{print}(\text{sum})]\rrbracket)]$

- $[(\llbracket [], \perp, \text{fun}(g, y, \text{sum}(y, 1))\rrbracket; \text{y3}, [\text{print}(\text{sum})]); (\llbracket [], \perp, [\text{apply}(\text{fun}(f, x, x), 2); \text{print}(\text{sum})]\rrbracket)]$
- $[(\llbracket 4, 3\rrbracket, \perp, \text{fun}(g, y, \text{sum}(y, 1))\rrbracket; \text{y3}, [\text{print}(\text{sum})]); (\llbracket [], \perp, [\text{apply}(\text{fun}(f, x, x), 2); \text{print}(\text{sum})]\rrbracket)]$
- $[(\llbracket 5\rrbracket, \perp, \text{fun}(g, y, \text{sum}(y, 1))\rrbracket; \text{y3}, [\text{print}(\text{sum})]); (\llbracket [], \perp, [\text{apply}(\text{fun}(f, x, x), 2); \text{print}(\text{sum})]\rrbracket)]$
- $[(\llbracket 5\rrbracket, \perp, [\text{apply}(\text{fun}(f, x, x), 2); \text{print}(\text{sum})]\rrbracket)]$
- $[(\llbracket 5\rrbracket, \perp, [\text{fun}(f, x, x); 2; \text{app}; \text{print}(\text{sum})]\rrbracket)]$
- $[(\llbracket 2; \text{fun}(f, x, x); 5\rrbracket, \perp, [\text{app}; \text{print}(\text{sum})]\rrbracket)]$
- $[(\llbracket 3, \perp, \text{fun}(f, x, x)\rrbracket; \text{x3}, [\text{x}]); (\llbracket 5\rrbracket, \perp, [\text{print}(\text{sum})]\rrbracket)]$
- $[(\llbracket 4(x), \perp, \text{fun}(f, x, x)\rrbracket; \text{x3}, [\text{x}]); (\llbracket 5\rrbracket, \perp, [\text{print}(\text{sum})]\rrbracket)]$
- $[(\llbracket 2, 3\rrbracket, \perp, \text{fun}(f, x, x)\rrbracket; \text{x3}, [\text{x}]); (\llbracket 5\rrbracket, \perp, [\text{print}(\text{sum})]\rrbracket)]$
- $[(\llbracket 2, 5\rrbracket, \perp, [\text{print}(\text{sum})]\rrbracket)]$
- $[(\llbracket 7\rrbracket, \perp, [\text{x}]\rrbracket)]$

(b) $\{([], \perp, [\text{apply}(\text{fun}(f, x, x), \text{fun}(h, z, \text{sum}(3, z))])\}$

$[(\llbracket [], \perp, [\text{apply}(\text{fun}(f, x, x), \text{fun}(h, z, \text{sum}(3, z))]\rrbracket)]$
 $[(\llbracket 3, \perp, [\text{fun}(f, x, x), \text{fun}(h, z, \text{sum}(3, z)), \text{app}]\rrbracket)]$
 $[(\llbracket \text{fun}(f, x, x), \perp, [\text{fun}(h, z, \text{sum}(3, z)), \text{app}]\rrbracket)]$
 $[(\llbracket \text{fun}(h, z, \text{sum}(3, z)); \text{fun}(f, x, x)\rrbracket, \perp, [\text{app}]\rrbracket] \quad \text{SECD} \quad [(\text{sec}); (\text{sec}); \dots]$
 $[(\llbracket 3, \perp, \text{fun}(f, x, x)\rrbracket; \text{fun}(h, z, \text{sum}(3, z))\rrbracket, \text{x3}, [\text{x}]); (\llbracket [], \perp, [\text{x}]\rrbracket)]$
 $[(\llbracket (x)\rrbracket, \perp, \text{fun}(f, x, x)\rrbracket; \text{fun}(h, z, \text{sum}(3, z))\rrbracket, \text{x3}, [\text{x}]); (\llbracket [], \perp, [\text{x}]\rrbracket)]$
 $[(\llbracket \text{fun}(h, \dots)\rrbracket, \perp, \text{fun}(f, x, x)\rrbracket; \text{fun}(h, z, \text{sum}(3, z))\rrbracket, \text{x3}, [\text{x}]); (\llbracket [], \perp, [\text{x}]\rrbracket)]$
 $[(\llbracket \text{fun}(h, z, \text{sum}(3, z))\rrbracket, \perp, [\text{x}]\rrbracket)]$

(c) $[[[]], \perp, [sum(apply(fun(f, x, apply(x, 4)), fun(h, z, sum(z, z))), 3)]]]$

$fun(f, x, apply(x, 4))$ $fun(h, z, sum(z, z))$

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$[[[]], \perp, [sum(apply(fun(f, x, apply(x, 4)), fun(h, z, sum(z, z))), 3)]]]$
 $[[[]], \perp, [apply(fun(f, x, apply(x, 4)), fun(h, z, sum(z, z))), 3; sum]]]$
 $[[[]], \perp, [fun(f, x, apply(x, 4)); fun(h, z, sum(z, z)); APP; 3; sum]]]$
 $[[[]], \perp, [fun(h, z, sum(z, z)); fun(f, x, apply(x, 4))], \perp, [APP; 3; sum]]]$
 $[[[]], \perp, [fun(f, x, apply(x, 4)) \text{ / } \text{f}, fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [apply(x, 4)]] : [[[]], \perp, [3; sum]]]$
 $[[[]], \perp, [fun(f, x, apply(x, 4)) \text{ / } \text{f}, fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [x, 4, APP]] : [[[]], \perp, [3; sum]]]$
 $[[[]], \perp, [fun(f, x, apply(x, 4)) \text{ / } \text{f}, fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [APP]] : [[[]], \perp, [3; sum]]]$
 $[[[]], \perp, [fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [sum(z, z)]] : [[[]], \perp, [fun(f, x, apply(x, 4)) \text{ / } \text{f}, fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [3]] : [[[]], \perp, [3; sum]]]$
 $[[[]], \perp, [fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [3, z, PRIM(sum)]] : [[[]], \perp, [fun(f, x, apply(x, 4)) \text{ / } \text{f}, fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [3]] : [[[]], \perp, [3; sum]]]$
 $[[[]], \perp, [fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [PRIM(sum)]] : [[[]], \perp, [fun(f, x, apply(x, 4)) \text{ / } \text{f}, fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [3]] : [[[]], \perp, [3; sum]]]$
 $[[[]], \perp, [fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [3]] : [[[]], \perp, [fun(f, x, apply(x, 4)) \text{ / } \text{f}, fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [3]] : [[[]], \perp, [3; sum]]]$
 $[[[]], \perp, [fun(f, x, apply(x, 4)) \text{ / } \text{f}, fun(h, z, sum(z, z)) \text{ / } \text{h, z}, [3]] : [[[]], \perp, [3; sum]]]$
 $[[[]], \perp, [3; sum]]]$
 $[[3; 7], \perp, [PRIM(sum)]]]$
 $[[10], \perp, [3]]]$

(d) $[[[]], \perp, [sum(apply(fun(f, x, sum(x, 3)), 5), apply(fun(g, y, sum(4, y)), 1)]]]$

$[[[]], \perp, [sum(apply(fun(f, x, sum(x, 3)), 5), apply(fun(g, y, sum(4, y)), 1)]]]$
 $[[[]], \perp, [apply(fun(f, x, sum(x, 3)), 5), apply(fun(g, y, sum(4, y)), 1), PRIM(sum)]]]$
 $[[[]], \perp, [fun(f, x, sum(x, 3)), 5, APP, apply(fun(g, y, sum(4, y)), 1), PRIM(sum)]]]$
 $[[[]], \perp, [fun(f, x, sum(x, 3)) \text{ / } \text{f}, 5, APP, apply(fun(g, y, sum(4, y)), 1), PRIM(sum)]]]$
 $[[[]], \perp, [fun(f, x, sum(x, 3)) \text{ / } \text{f}, 5, [sum(x, 3)]] ; [[[]], \perp, [apply(fun(g, y, sum(4, y)), 1), PRIM(sum)]]]$
 $[[[]], \perp, [fun(f, x, sum(x, 3)) \text{ / } \text{f}, 5, [3]] ; [[[]], \perp, [apply(fun(g, y, sum(4, y)), 1), PRIM(sum)]]]$
 $[[[]], \perp, [apply(fun(g, y, sum(4, y)), 1), PRIM(sum)]]]$
 $[[[]], \perp, [apply(fun(g, y, sum(4, y)), 1), PRIM(sum)]]]$
 $[[[]], \perp, [fun(g, y, sum(4, y)) \text{ / } \text{g, y}, 1, [APP, PRIM(sum)]]]$
 $[[[]], \perp, [fun(g, y, sum(4, y)) \text{ / } \text{g, y}, 1, [APP, PRIM(sum)]]]$
 $[[[]], \perp, [fun(g, y, sum(4, y)) \text{ / } \text{g, y}, 1, [sum(4, y)]] ; [[[]], \perp, [PRIM(sum)]]]$
 $[[[]], \perp, [sum(4, y) \text{ / } \text{4, y}, 1, [3]] ; [[[]], \perp, [PRIM(sum)]]]$
 $[[[]], \perp, [PRIM(sum)]]]$
 $[[[]], \perp, [PRIM(sum)]]]$

(e) $[[[]], \perp, [minus(apply(fun(f, x, sum(x, x)), 8), apply(fun(f, x, sum(x, 5)), 1)]]]$

$[[[]], \perp, [minus(apply(fun(f, x, sum(x, x)), 8), apply(fun(f, x, sum(x, 5)), 1)]]]$
 $[[[]], \perp, [apply(fun(f, x, sum(x, x)), 8); apply(fun(f, x, sum(x, 5)), 1); PRIM(minus)]]]$
 $[[[]], \perp, [fun(f, x, sum(x, x)); 8; APP; apply(fun(f, x, sum(x, 5)), 1); PRIM(minus)]]]$
 $[[[]], \perp, [fun(f, x, sum(x, x)) \text{ / } \text{f, x}, 8; APP; apply(fun(f, x, sum(x, 5)), 1); PRIM(minus)]]]$
 $[[[]], \perp, [fun(f, x, sum(x, x)) \text{ / } \text{f, x}, 8, [sum(x, x)]] ; [[[]], \perp, [apply(fun(f, x, sum(x, 5)), 1); PRIM(minus)]]]$
 $[[[]], \perp, [fun(f, x, sum(x, x)) \text{ / } \text{f, x}, 8, [3]] ; [[[]], \perp, [apply(fun(f, x, sum(x, 5)), 1); PRIM(minus)]]]$
 $[[[]], \perp, [fun(f, x, sum(x, x)) \text{ / } \text{f, x}, 8, [3]] ; [[[]], \perp, [PRIM(minus)]]]$

$$\begin{aligned}
& [([], \perp, \underbrace{\text{FUN}(f, x, \text{SUM}(x, 5))}_{E_1}; \underbrace{8}_{t_1}, \underbrace{[\text{SUM}(x, 5)]}_{t_1}); ([], \perp, \underbrace{[\text{APPLY}(\text{FUN}(f, x, \text{SUM}(x, 5)), 1)]}_{t_1}; \text{PRIM}(\text{MINUS}))]] \\
& [([], \perp, \underbrace{E_1}_{t_1}, [x, x, \text{PRIM}(\text{SUM})]); t_1] \\
& [([8; 5], \perp, \underbrace{E_2}_{t_1}, [\text{PRIM}(\text{SUM})]]); t_1] \\
& [([16], \perp, \underbrace{E_1}_{t_1}, []]); t_1] \\
& [([16], \perp, \underbrace{[\text{FUN}(f, x, \text{SUM}(x, 5))]}_{t_1}; 1; \text{APP}; \text{PRIM}(\text{MINUS}))]] \\
& [([\text{FUN}(f, x, \text{SUM}(x, 5))]; 1; 16], \perp, \underbrace{[\text{APP}; \text{PRIM}(\text{MINUS})]}_{t_2})] \\
& [([], \perp, \underbrace{\text{FUN}(f, x, \text{SUM}(x, 5))}_{E_2}; \underbrace{1}_{t_2}, \underbrace{[\text{SUM}(x, 5)]}_{t_2}); ([16], \perp, \underbrace{[\text{PRIM}(\text{MINUS})]}_{t_2})]]
\end{aligned}$$

$$\begin{aligned}
& [([], \perp, \underbrace{E_2}_{t_2}, [x, 5, \text{PRIM}(\text{SUM})]); t_2] \\
& [([5; 1], \perp, \underbrace{E_2}_{t_2}, [\text{PRIM}(\text{SUM})]]); t_2] \\
& [([6], \perp, \underbrace{E_2}_{t_2}, []]); t_2] \\
& [([6; 16], \perp, \underbrace{[\text{PRIM}(\text{MINUS})]}_{t_2})] \\
& [([10], \perp, \underbrace{[]}_{t_2})]
\end{aligned}$$

(f) $[[[], \perp, [\text{apply}(\text{fun}(f, x, \text{apply}(x, \text{apply}(\text{fun}(g, y, \text{sum}(y, 2)), 1))), \text{fun}(h, z, \text{sum}(3, z))]]]]]$

$$\begin{aligned}
& [([], \perp, \underbrace{[\text{APPLY}(\text{FUN}(f, x, \underbrace{\text{APPLY}(\text{FUN}(g, y, \text{SUM}(y, 2)), 1)})_{E_1}, \text{FUN}(h, z, \text{SUM}(3, z))]}_{t_1})]_{t_1})]] \\
& [([], \perp, \underbrace{[\text{FUN}(f, x, \underbrace{\text{APPLY}(\text{FUN}(g, y, \text{SUM}(y, 2)), 1)})_{E_1}, \text{FUN}(h, z, \text{SUM}(3, z))]}_{t_1}); \text{APP}]]] \\
& [([\text{FUN}(h, z, \text{SUM}(3, z))]; \text{FUN}(f, x, \underbrace{\text{APPLY}(\text{FUN}(g, y, \text{SUM}(y, 2)), 1)})_{t_1}, \perp, \underbrace{[\text{APP}]}_{t_1})]] \\
& [([], \perp, \underbrace{\text{FUN}(f, x, \underbrace{\text{APPLY}(\text{FUN}(g, y, \text{SUM}(y, 2)), 1)})_{E_1}, \text{FUN}(h, z, \text{SUM}(3, z))}_{t_1}); \underbrace{[\text{APPLY}(\text{FUN}(g, y, \text{SUM}(y, 2)), 1)]}_{t_1}); ([], \perp, \underbrace{[]}_{t_1})]] \\
& [([], \perp, \underbrace{E_1}_{t_1}, [x; \text{APPLY}(\text{FUN}(g, y, \text{SUM}(y, 2)), 1)]; \text{APP}]); ([], \perp, \underbrace{[]}_{t_1})]] \\
& [([\text{FUN}(h, z, \text{SUM}(3, z))]; \perp, \underbrace{E_1}_{t_1}, [\text{APPLY}(\text{FUN}(g, y, \text{SUM}(y, 2)), 1)]; \text{APP}]); ([], \perp, \underbrace{[]}_{t_1})]] \\
& \text{FUN}(h, z, \text{SUM}(3, z))
\end{aligned}$$

$$[([\text{FUN}(h, z, \text{SUM}(3, z))]; \perp, \underbrace{E_1}_{t_1}, [\text{FUN}(g, y, \text{SUM}(y, 2)); 1; \text{APP}; \text{APP}]); ([], \perp, \underbrace{[]}_{t_1})]]$$

$$\begin{aligned}
& [([1; \text{FUN}(g, y, \text{SUM}(y, 2))]; \text{FUN}(h, z, \text{SUM}(3, z)), \perp, \underbrace{E_1}_{t_1}, [\text{APP}; \text{APP}]); ([], \perp, \underbrace{[]}_{t_1})]] \\
& [([1]; \perp, \underbrace{\text{FUN}(g, y, \text{SUM}(y, 2))}_{E_2}; \underbrace{1}_{t_2}, \underbrace{[\text{SUM}(y, 2)]}_{t_2}); ([\text{FUN}(h, z, \text{SUM}(3, z))]; \perp, \underbrace{E_1}_{t_1}, \underbrace{[]}_{t_1})]] \\
& \text{FUN}(h, z, \text{SUM}(3, z))
\end{aligned}$$

$$[([], \perp, \underbrace{E_2}_{t_2}, [y; 2; \text{PRIM}(\text{SUM})]); D_1]$$

$$[([2; 1], \perp, \underbrace{E_2}_{t_2}, [\text{PRIM}(\text{SUM})]]); D_1]$$

$$[([3], \perp, \underbrace{E_2}_{t_2}, []]); D_1]$$

$$[([3; \text{FUN}(h, z, \text{SUM}(3, z))]; \perp, \underbrace{E_1}_{t_1}, [\text{APP}]); ([], \perp, \underbrace{[]}_{t_1})]]$$

$$[([], \perp, \underbrace{\text{FUN}(h, z, \text{SUM}(3, z))}_{E_3}, \underbrace{3}_{t_3}, \underbrace{[\text{SUM}(3, z)]}_{t_3}); ([], \perp, \underbrace{[]}_{t_1})]]$$

$$[([], \perp, \underbrace{E_3}_{t_3}, [3; 2; \text{PRIM}(\text{SUM})]); ([], \perp, \underbrace{[]}_{t_1})]]$$

$$[([3; 3], \perp, \underbrace{E_3}_{t_3}, [\text{PRIM}(\text{SUM})]]); ([], \perp, \underbrace{[]}_{t_1})]]$$

$$[([6; 3], \perp, \underbrace{E_3}_{t_3}, []]); ([], \perp, \underbrace{[]}_{t_1})]]$$

$$[([6; 3], \perp, \underbrace{E_3}_{t_3}, []]); ([], \perp, \underbrace{[]}_{t_1})]]$$

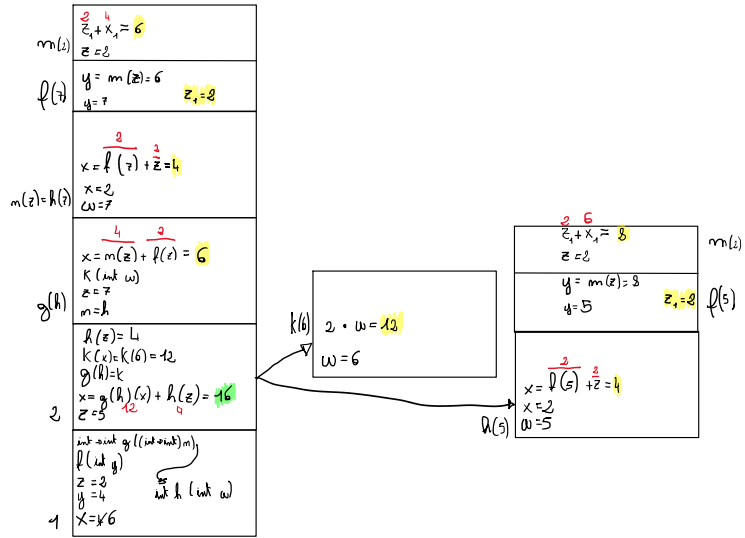
$$[([6; 3], \perp, \underbrace{[]}_{t_1})]]$$

ESERCIZIO 15

```

01 int x = 4;
02 int y = 4;
03 int z = 2;
04 int f (int y) {
05     int m (int z) {
06         return z + x;
07     }
08     return z;
09 }
10 (int -> int) g ((int -> int) m) {
11     int k (int w) {
12         return 2 * w;
13     }
14     return k;
15 }
16 int h (int w) {
17     int x = 2;
18     x = f(w) + z;
19     return x;
20 }
21 int z = 5;
22 x = g(h(x) + h(z));

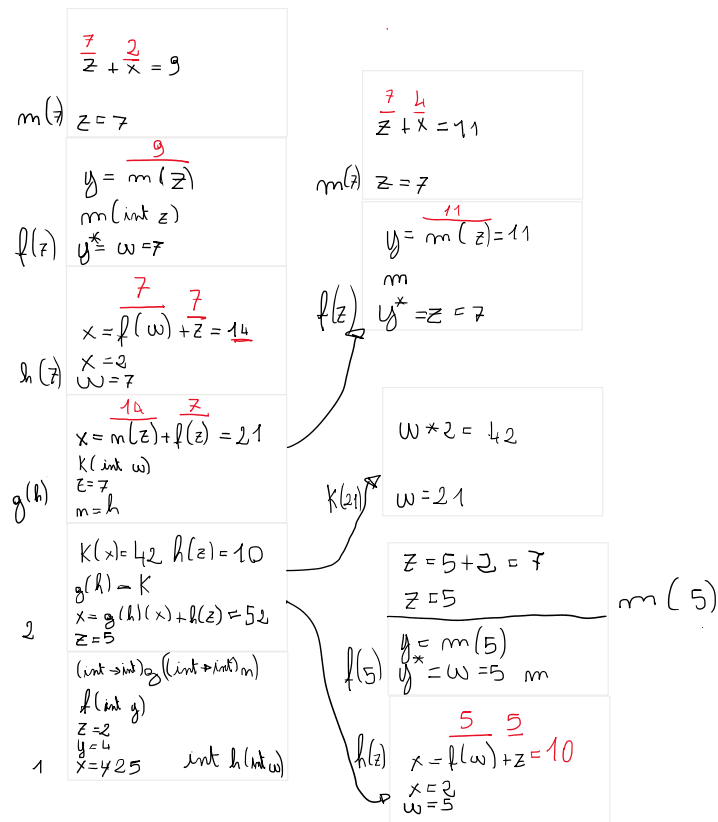
```

(a) che valore contiene la variabile x alla linea 20 se lo scope è statico (Risultato: 16)

```

01 {int x = 4;
02 int y = 4;
03 int z = 2;
04 int f (int y) {
05     int m (int z) {
06         return z + x;
07     }
08     return z;
09 }
10 (int -> int) g ((int -> int) m) {
11     int k (int w) {
12         return 2 * w;
13     }
14     return k;
15 }
16 int h (int w) {
17     int x = 2;
18     x = f(w) + z;
19     return x;
20 }
21 int z = 5;
22 x = g(h(x) + h(z));

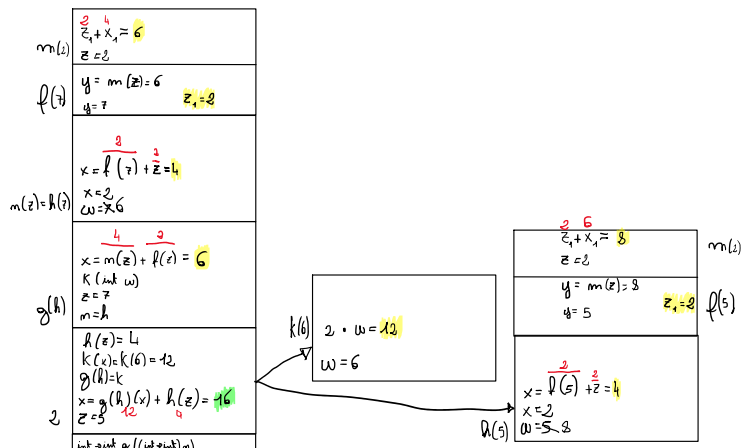
```

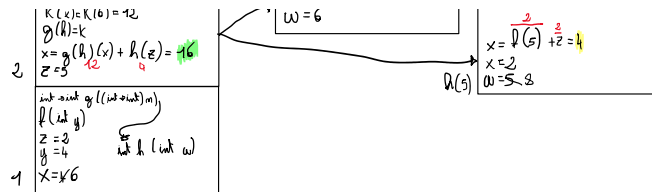
(b) che valore contiene la variabile x alla linea 20 se lo scope è dinamico e la politica è quella del shallow binding (Risultato: 52)(c) che valore contiene la variabile x alla linea 20 se lo scope è statico e il parametro y della funzione f è passato per riferimento (Risultato: 16)

```

01 {int x = 4;
02 int y = 4;
03 int z = 2;
04 int f (int y) {
05     int m (int z) {
06         return z + x;
07     }
08     return z;
09 }
10 (int -> int) g ((int -> int) m) {
11     int k (int w) {
12         return 2 * w;
13     }
14     return k;
15 }
16 int h (int w) {
17     int x = 2;
18     x = f(w) + z;
19     return x;
20 }
21 int z = 5;
22 x = g(h(x) + h(z));

```





(d) che valore contiene la variabile x alla linea 20 se lo scope è dinamico, la politica è quella dello shallow binding e il parametro y della funzione f è passato per riferimento (Risultato: **60**)

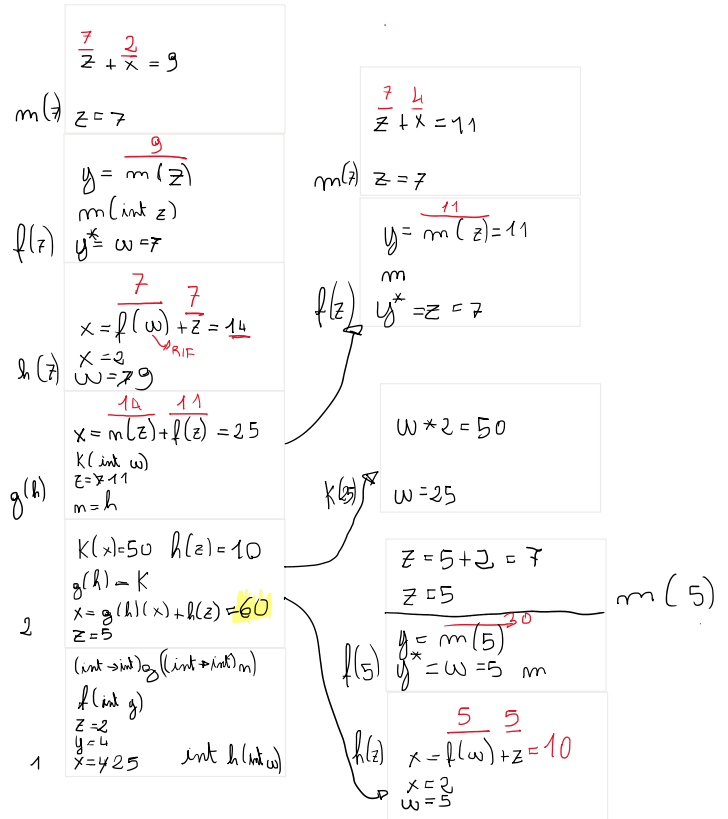
```

01 int x = 4;
02 int y = 4;
03 int z = 2;
04 int f (int y) {
05     int m (int z) {
06         return z + x;
07     }
08     return z;
09 }
10 (int -> int) g ((int -> int) n) {
11     int k (int w) {
12         return 2 * w;
13     }
14     return k;
15 }
16 int h (int w) {
17     int x = 2;
18     x = f(w) + z;
19     return x;
20 }
21 x = g(h)(x) + h(z);

```

(e) che valore è associato alla variabile x nella funzione h quando viene chiamata alla linea 20 (Risultato: **6**)

NON CAPISCO PERCHE' 6, IMMAGINANDO CHE SI RIFERISCA AD $h(z)$ IL VALORE E' 25

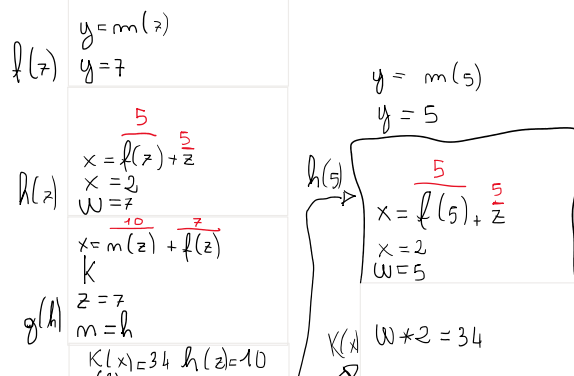


(f) che valore contiene la variabile x alla linea 20 se lo scope è dinamico e la politica è quella del deep binding

```

01 int x = 4;
02 int y = 4;
03 int z = 2;
04 int f (int y) {
05     int m (int z) {
06         return z + x;
07     }
08     return z;
09 }
10 (int -> int) g ((int -> int) n) {
11     int k (int w) {
12         return 2 * w;
13     }
14     return k;
15 }
16 int h (int w) {
17     int x = 2;
18     x = f(w) + z;
19     return x;
20 }
21 x = g(h)(x) + h(z);

```



ESERCIZIO 16

```

01 {int x = 1;
02 int y = 1;
03 int z = 2;
04 int f (int y) {
05     y = x + z;
06     return y;
07 int m (int x) {;
08     return y + x;
09 int h1 (int w) {
10     int x = 3;
11     x = f(x) * w;
12     return x;
13 int h2 (int x) {
14     int w = 1;
15     w = f(x) + w;
16     return w;
17 int -> int g (int -> int n) {
18     int k (int w) {
19         return m(w) + 1;
20     int x = 2;
21     int z = 3;
22     y = n(x) + f(x);
23     return k;
24 {int x = 2;
25     int z = 2;
26     x = g(h1)(x) + h2(y) + x;

```

(a) che valore contiene la variabile x alla linea 26 se lo scope è statico (Risultato: 14)

```

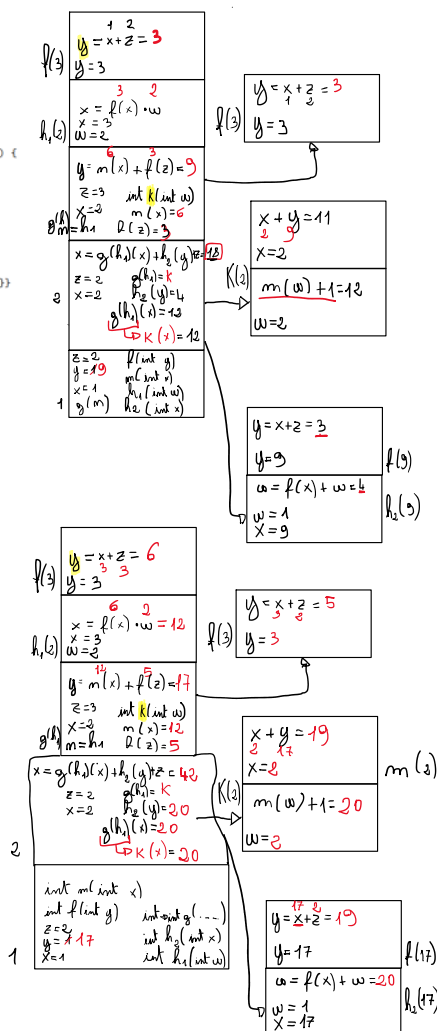
01 {int x = 1;
02 int y = 1;
03 int z = 2;
04 int f (int y) {
05     y = x + z;
06     return y;
07 int m (int x) {;
08     return y + x;
09 int h1 (int w) {
10     int x = 3;
11     x = f(x) * w;
12     return x;
13 int h2 (int x) {
14     int w = 1;
15     w = f(x) + w;
16     return w;
17 int -> int g (int -> int n) {
18     int k (int w) {
19         return m(w) + 1;
20     int x = 2;
21     int z = 3;
22     y = n(x) + f(x);
23     return k;
24 {int x = 2;
25     int z = 2;
26     x = g(h1)(x) + h2(y) + x;

```

(b) che valore contiene la variabile x alla linea 26 se lo scope è dinamico o la politica è quella del shallow binding (Risultato: 24)

$g(h)$
 $z = 7$
 $m = h$
 $K(x) = 34$
 $h(z) = 10$
 $g(h) = K$
 $x = g(h)(x) + h(z) = 44$
 $z = 5$
 g
 h
 f
 $z = 2$
 $y = 4$
 $x = 17$

$w * 2 = 34$
 $w = 17$



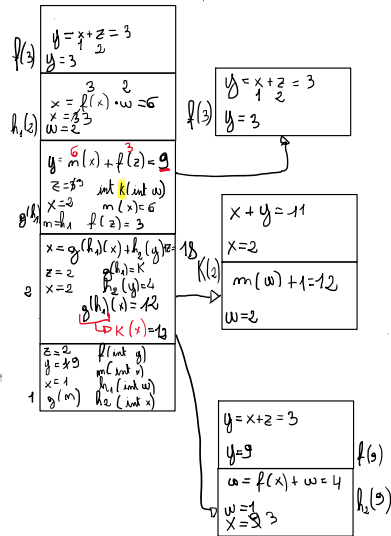
$y = x + z = 3$

```

01 {int x = 1;
02   int y = 1;
03   int z = 2;
04   int f (int y) {
05     y = x + z;
06     return y;
07   }
08   int m (int x) {
09     return y + x;
10   }
11   int h1 (int w) {
12     int x = 3;
13     x = f(x) * w;
14     return x;
15   }
16   int w = 1;
17   w = f(x) + w;
18   return w;
19 }
20 int -> int g (int -> int n) {
21   int k (int w) {
22     return m(w) + 1;
23   }
24   int x = 2;
25   int z = 3;
26   y = n(x) + f(z);
27   return k;
28 }
29 {int x = 2;
30   int z = 2;
31   x = g(h1)(x) + h2(y) + z;
32 }

```

(c) che valore contiene la variabile x alla linea 26 se lo scope è statico e il parametro y della funzione f è passato per riferimento (Risultato: **14**)



(d) che valore contiene la variabile x alla linea 26 se lo scope è dinamico, la politica è quella dello shallow binding e il parametro y della funzione f è passato per riferimento

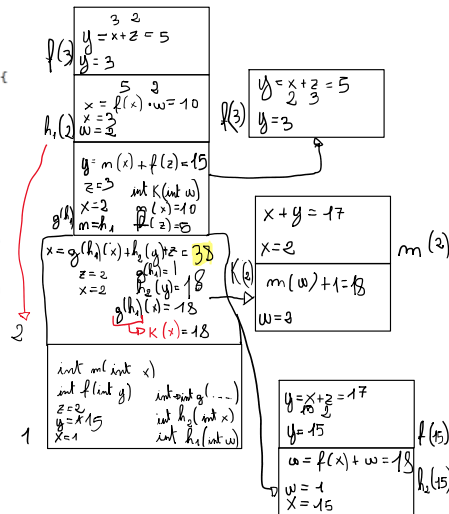
(e) che valore è associato alla variabile x nella funzione $h2$ quando viene chiamata alla linea 26

```

01 {int x = 1;
02   int y = 1;
03   int z = 2;
04   int f (int y) {
05     y = x + z;
06     return y;
07   }
08   int m (int x) {
09     return y + x;
10   }
11   int h1 (int w) {
12     int x = 3;
13     x = f(x) * w;
14     return x;
15   }
16   int w = 1;
17   w = f(x) + w;
18   return w;
19 }
20 int -> int g (int -> int n) {
21   int k (int w) {
22     return m(w) + 1;
23   }
24   int x = 2;
25   int z = 3;
26   y = n(x) + f(z);
27   return k;
28 }
29 {int x = 2;
30   int z = 2;
31   x = g(h1)(x) + h2(y) + z;
32 }

```

(f) che valore contiene la variabile x alla linea 26 se lo scope è dinamico e la politica è quella del deep binding (Risultato: **22**)

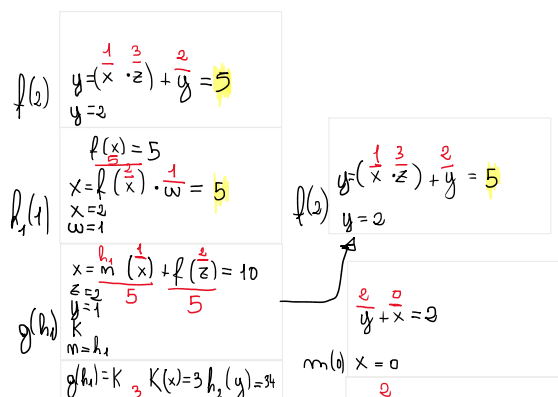


ESERCIZIO 17

```

01 {int x = 1;
02   int y = 2;
03   int z = 3;
04   int f (int y) {
05     y = (x * z) + y;
06     return y;
07   }
08   int m (int x) {
09     return y + x;
10   }
11   int h1 (int w) {
12     int x = 2;
13     x = f(x) * w;
14     return x;
15   }
16   int w = 2;
17   w = f(x) + w;
18   return w;
19 }
20 int -> int g (int -> int n) {
21   int k (int w) {
22     return m(w) + 1;
23   }
24   int y = 1;
25   int z = 2;
26   x = n(x) + f(z);
27   return k;
28 }
29 {int x = 0;
30   int z = 1;
31   x = g(h1)(x) + h2(y) + y;
32 }

```

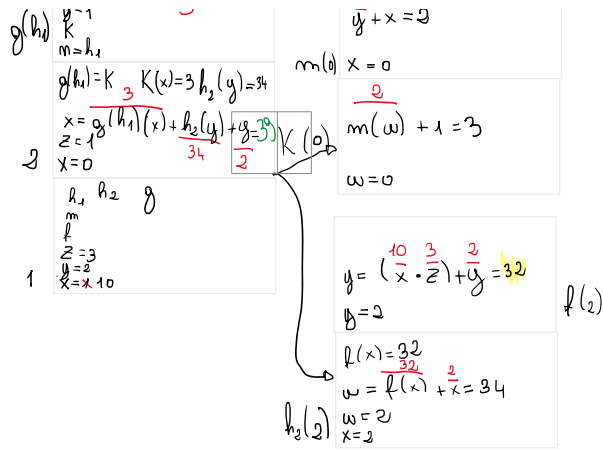


```

09 int h1(int w) {
10     int x = 2;
11     x = f(x) * w;
12     return x;
13 int h2(int x) {
22     x = n(x) + f(x);
23     return k;
24 {int x = 0;
25     int z = 1;
26     x = g(h1)(x) + h2(y) + y;}}

```

(a) che valore contiene la variabile x alla linea 26 se lo scope è statico (Risultato: 39)

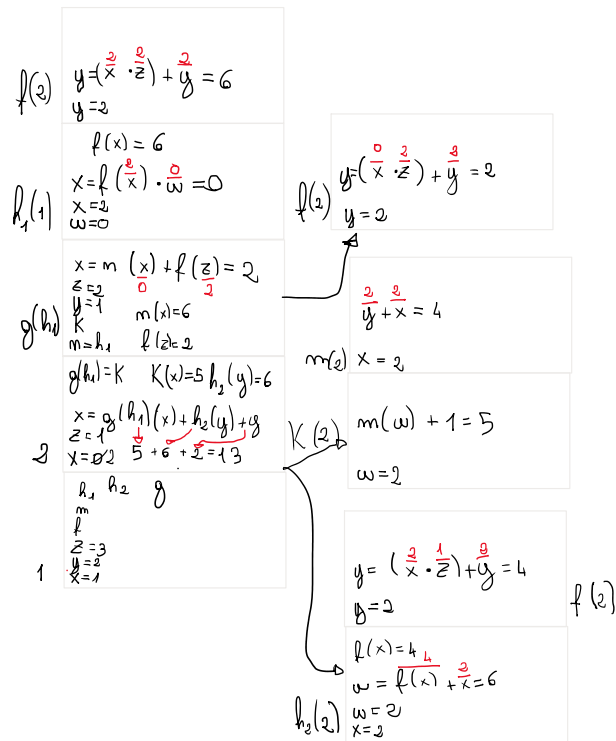


```

01 {int x = 1;
02 int y = 2;
03 int z = 3;
04 int f(int y) {
05     y = (x * z) * y;
06     return y;
07 int m(int x) {;
08     return y + x;
09 int h1(int w) {
10     int x = 2;
11     x = f(x) * w;
12     return x;
13 int h2(int x) {
14     int w = 2;
15     w = f(x) * x;
16     return w;
17 int -> int g(int -> int n) {
18     int k(int w) {
19         return m(w) + 1;
20     int y = 1;
21     int z = 2;
22     x = n(x) + f(x);
23     return k;
24 {int x = 0;
25     int z = 1;
26     x = g(h1)(x) + h2(y) + y;}}

```

(b) che valore contiene la variabile x alla linea 26 se lo scope è dinamico e la politica è quella del shallow binding (Risultato: 13)

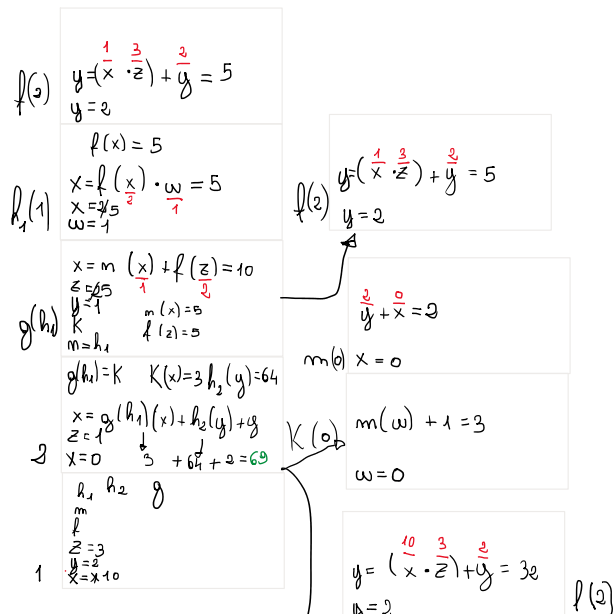


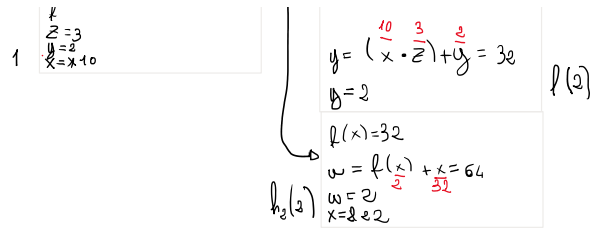
(c) che valore contiene la variabile x alla linea 26 se lo scope è statico e il parametro y della funzione f è passato per riferimento (Risultato: 69)

```

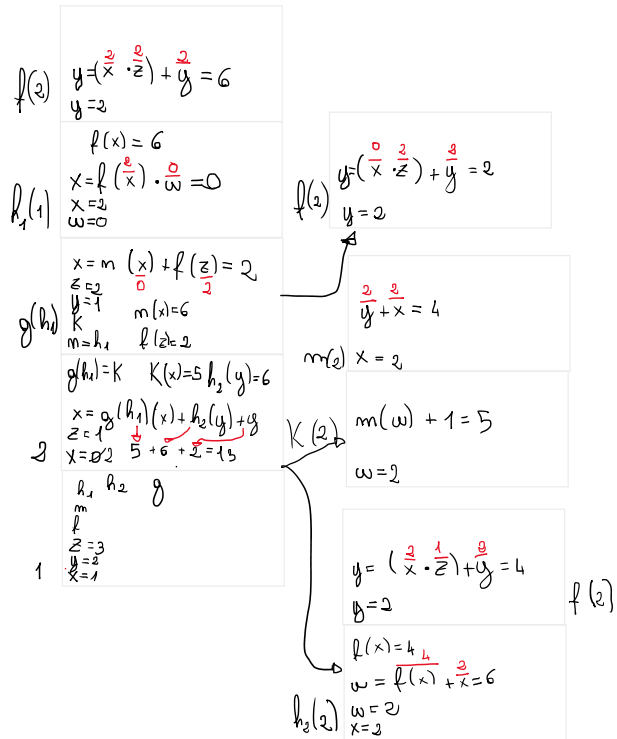
01 {int x = 1;
02 int y = 2;
03 int z = 3;
04 int f(int y) {
05     y = (x * z) * y;
06     return y;
07 int m(int x) {;
08     return y + x;
09 int h1(int w) {
10     int x = 2;
11     x = f(x) * w;
12     return x;
13 int h2(int x) {
14     int w = 2;
15     w = f(x) * x;
16     return w;
17 int -> int g(int -> int n) {
18     int k(int w) {
19         return m(w) + 1;
20     int y = 1;
21     int z = 2;
22     x = n(x) + f(x);
23     return k;
24 {int x = 0;
25     int z = 1;
26     x = g(h1)(x) + h2(y) + y;}}

```





(d) che valore contiene la variabile x alla linea 26 se lo scope è dinamico, la politica è quella dello shallow binding e il parametro y della funzione f è passato per riferimento (Risultato: 15)



(e) che valore è associato alla variabile x nella funzione h_2 quando viene chiamata alla linea 26 $\rightarrow x=2$

IL PROF HA MESSO 0 COME RISPOSTA, MA MI SEMBRA IMPOSSIBILE

(f) che valore contiene la variabile x alla linea 26 se lo scope è dinamico e la politica è quella del deep binding (Risultato: 13)

